A NISQ-Deployable Quantum Orthogonal Attention Circuit for Dark Matter Classification: Matching, Not Beating, Classical Attention, and a Cautionary Tale About Training Instability

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Abstract

We introduce QOVT, a Quantum Orthogonal Vision Transformer whose attention matrices are produced by a genuine, NISQ-deployable 6-qubit gate-model quantum circuit – an RY+CNOT ring with L_{circ}=8 layers and only T=384 free angles. On strong-gravitational-lensing images from three simulated dark-matter models, this circuit matches the held-out AUC of a standard 8-head classical multi-head-attention baseline (T=65,536) at every training-set size we test, from N=500 to the full $\sim$25k images per class, while using 170× fewer attention parameters. A pre-registered, held-out-test, 10-seed evaluation (N=500/class) further shows the circuit significantly beats an unconstrained (non-orthogonal) baseline on one of two datasets (Model_II: Δ=+15.7pp, Holm-corrected p=0.0039, 10/10 seeds) but not the other (Model_III: Δ=+7.4pp, p=0.0137, not significant) – a directional, dataset-dependent, unreplicated finding, not a general claim of quantum advantage. As a companion ablation we also study Givens-butterfly decomposition, a *classical* (non-quantum, no gate-model circuit at all) construction of the same 64×64 orthogonal matrix with 4× more parameters (T=1,536); it fails catastrophically and non-monotonically in about half of all runs, including at 5× more training data, showing that classical constructions of "the same" matrix do not inherit the quantum circuit's stability. We use these results to empirically test the Caro et al. generalization bound as a circuit-design criterion and find it does not predict our observations: a circuit-depth ablation shows accuracy *increasing*, not decreasing, with more circuit parameters at low N. All code, pre-registration documents, raw results, and this manuscript's source are public.

1 Introduction

Strong gravitational lensing is one of the few direct observational probes of dark-matter substructure: the way a foreground galaxy bends light from a background source encodes the mass distribution along the line of sight, including sub-halos too faint or too dark to see directly [7]. Different dark-matter models predict qualitatively different substructure – cold dark matter (CDM) produces discrete, point-mass-like sub-halos, while ultralight axion (vortex) dark matter produces extended, wave-like density modulations [13] – so classifying a lensing image by which substructure model it is consistent with is, in effect, a machine-learning test of fundamental physics. This makes the classification task scientifically consequential in a way that goes beyond a benchmark: getting it right, and getting it right with as few labeled (typically simulation-generated, and expensive to produce at fidelity) images as possible, directly bears on which dark-matter models next-generation surveys can discriminate between.

Classical deep learning has driven essentially all recent progress on this task. Convolutional networks were the first to show that lens-finding and substructure classification could be automated at scale [12, 14], and CNN-based morphology classifiers remain strong baselines for the CDM-vs-axion-vs-none problem specifically [1]. Simulation-based inference extends this to full posterior estimation over sub-halo population properties rather than a single label [2]. More recently, the field has moved to attention-based architectures: Vision Transformers [8] and, specifically for this task, a masked-autoencoder-pretrained ViT that is the current published state of the art on DeepLense-style substructure classification [24]. Attention is powerful precisely because each layer learns a dense, quadratic-in-tokens set of interactions – but that same density means a standard multi-head-attention layer at our model's width (D=64, 8 heads) already spends 65,536 parameters on projections alone, per layer, raising the question of whether a structurally constrained – and potentially quantum-native – parameterization of attention can match this accuracy far more cheaply.

Quantum machine learning offers exactly such a structural constraint: unitary (here, real-orthogonal) circuits are parameter-efficient by construction, since a circuit with T free gate angles can only reach a T-dimensional sub-manifold of all possible attention matrices. This is formalized by the generalization bound of Caro et al. [3] (see also [16, 20]), which bounds the train/test gap by $O(\sqrt{T/N})$ and has motivated a family of quantum-attention transformers – Quantum Orthogonal Transformers built from Givens-rotation ("RBS") circuits [5], quantum attention for high-energy physics [29, 30, 6], and quantum self-attention for text [19] – alongside a broader line of work on orthogonal and unitary neural-network layers, both quantum [17, 18] and classically orthogonality-regularized [15, 9]. A parallel and largely independent line of quantum ML work uses quantum circuits as kernel or feature-map devices [11, 10] and has been applied to other high-energy-physics classification problems [28, 22]. Running any of this on real hardware, however, means confronting the practical limits of the current NISQ era [25, 26] and the well-documented trainability failures – barren plateaus and over-expressibility – that shallow, highly entangled circuits are prone to [21, 4, 27]. These constraints favor circuits that are not just parameter-efficient on paper but genuinely shallow and hardware-realizable, a distinction this paper takes seriously: several published "quantum orthogonal attention" constructions, including the Givens-butterfly variant we study as a control, parameterize an orthogonal matrix using an angle-counting scheme borrowed from circuit design, but do not correspond to any gate sequence that runs on a quantum device.

Our contributions are:

1. **A genuinely deployable quantum circuit.** We design and implement RY+CNOT, a 6-qubit, 8-layer, ring-connectivity gate-model circuit (96 gates total: 48 single-qubit RY + 48 two-qubit CNOT, T=384 free angles) that produces the 64×64 orthogonal attention matrices used by every QOVT layer, and show it matches an 8-head classical multi-head-attention baseline's held-out AUC (T=65,536, 170× more parameters) at every training-set size we test.
2. **A pre-registered statistical evaluation.** Correcting a methodological flaw (discovery and confirmatory samples overlapping) we found in an earlier campaign in this project, we pre-register hypotheses, a held-out test split, and a Holm-Bonferroni-corrected test battery *before* training, then run 10 seeds × 2 datasets × 5 arms at N=500/class. RY+CNOT significantly beats an unconstrained baseline on one of two datasets – a real but dataset-dependent, unreplicated, directional finding.
3. **An empirical test of the generalization-bound design rationale.** A circuit-depth ablation ($T \in \{192, 384, 576\}$) shows held-out accuracy *increasing* with more circuit parameters at low N, the opposite of what the Caro bound would predict if it were the operative mechanism – we still favor a shallow circuit, but for NISQ deployability, not for a generalization-bound advantage we do not observe.
4. **A cautionary companion control.** Givens-butterfly decomposition constructs the same class of orthogonal matrix *classically* (no quantum circuit exists for it), with 4× more parameters, and fails in roughly half of all runs, non-monotonically in N – evidence that the RY+CNOT circuit's stability is a property of the circuit, not of orthogonal attention in general.

2 Method

2.1 Preliminaries

Symbol	Meaning
x	input lensing image, 1×64×64
D=64	token embedding dimension
L=4	number of QO-attention layers in the backbone
N_Q=6	qubits in the RY+CNOT circuit ($2^{N_Q}=D$)
L_{circ}=8	RY+CNOT circuit layers (default)
$U,V \in O(D)$	orthogonal matrices used as query/value projections
T	number of free circuit / layer angles
N	training images per class

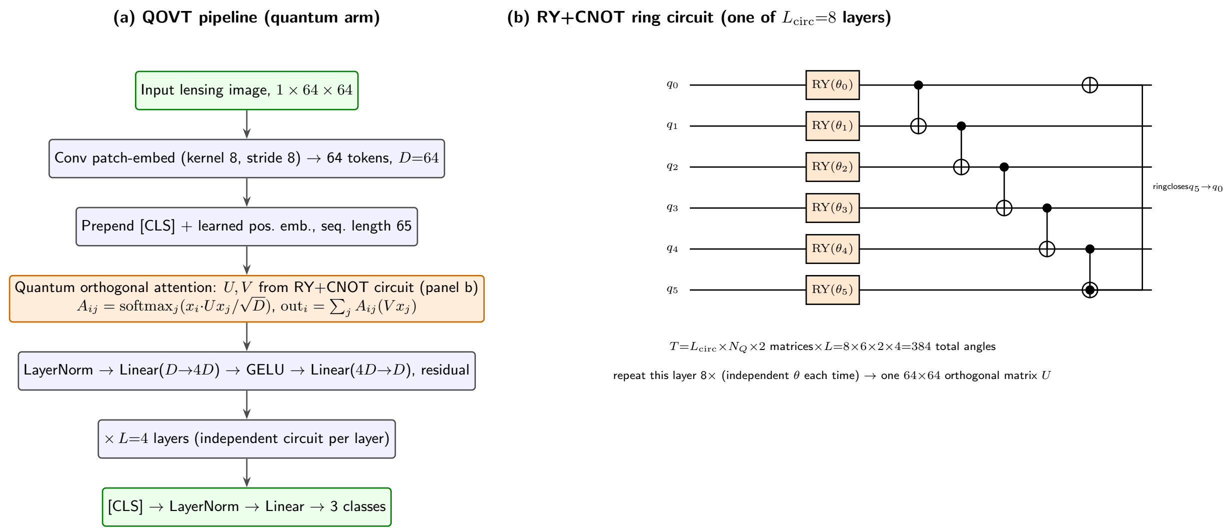

Figure 1: **The quantum arm of QOVT.** (a) End-to-end pipeline: only the quantum RY+CNOT circuit is shown as the source of U, V – classical controls (Givens butterfly, sham, classical MHA) are described separately in Section 2.5 and are not part of this architecture diagram. (b) The RY+CNOT ring circuit itself, one of the L_{circ}=8 repeated layers: an RY rotation on every qubit followed by a ring of nearest-neighbor CNOTs (including the wrap-around $q_5 \to q_0$ gate).

Every attention layer of QOVT needs two $D \times D$ orthogonal matrices, U and V, in place of the usual learned W_Q, W_K, W_V projections. A circuit (or, for controls, a classical parameterization) with T free scalar parameters can only reach a T-dimensional sub-manifold of $O(D)$. Caro et al. [3] show that for a hypothesis class with T trainable parameters trained on N i.i.d. samples, with high probability

$$\mathscr{R}(\hat{h}) - \hat{\mathscr{R}}(\hat{h}) \leq C\sqrt{T/N} + O\left(\sqrt{\frac{\log(1/\delta)}{N}}\right),$$

i.e. the train/test generalization gap shrinks with fewer parameters. This bound is the design rationale several quantum-attention papers cite for using small-T circuits [16, 20]; we adopt it only as a *heuristic* motivating our choice of a shallow circuit below, and we test it directly as an empirical claim in Section 3.4.

2.2 Overall Architecture (Quantum Arm)

Figure 1(a) shows the full quantum-arm pipeline. A $1 \times 64 \times 64$ lensing image is patch-embedded by a single stride-8 convolution into 64 tokens of dimension D=64; a learnable [CLS] token is prepended and a learned positional embedding added, giving a sequence of length 65. Each of L=4 QO-attention layers computes its own pair of orthogonal matrices U_ℓ, V_ℓ from an independently parameterized instance of the RY+CNOT circuit (Section 2.3), then applies an *asymmetric* orthogonal attention,

$$A_{ij} = \mathrm{softmax}_j\big(x_i \cdot Ux_j/\sqrt{D}\big), \qquad \mathrm{out}_i = \textstyle\sum_j A_{ij}\,(Vx_j),$$

i.e. queries are left unprojected (W_Q=I), keys use U, and values use V – distinct from the symmetric W_Q=W_K=U design used elsewhere in the literature (Section 5, Limitations). A residual connection, LayerNorm, and a standard $D \to 4D \to D$ GELU MLP (also residual) follow, and this block repeats L=4 times. The [CLS] token's final representation is normalized and linearly projected to the three class logits (axion / CDM / no-substructure).

2.3 The RY+CNOT Ring Circuit

Definition 1 (RY+CNOT ring circuit)**.** *On N_Q=6 qubits, one* layer *applies* RY(θ_q) *to every qubit $q \in \{0, \ldots, 5\}$, then a ring of CNOTs* $(0 \to 1), (1 \to 2), \ldots, (4 \to 5), (5 \to 0)$*. The full circuit stacks L_{circ}=8 such layers with independent angles,* *and its* $2^{N_Q}\times 2^{N_Q} = 64\times 64$ *unitary is read out as the orthogonal matrix* U *(a second, independently-parameterized copy of the same circuit family produces* V*). Total free angles per attention layer:* $T = L_{\text{circ}} \times N_Q \times 2$ *matrices* $= 8\times 6\times 2 = 96$*; across all* L=4 *attention layers,* $T_{\text{total}} = 384$.

Two properties make this circuit both a genuine quantum circuit and practical to run at scale. First, because RY and CNOT are both real-valued gates in the computational basis, the resulting unitary is *exactly* real – no complex amplitude ever arises – so its action is precisely a real 64×64 orthogonal matrix, exactly what the attention layer needs, with no approximation or discarded imaginary part. Second, the circuit is shallow and low-connectivity: 96 gates total (48 single-qubit RY, 48 two-qubit CNOT), nearest-neighbor ring connectivity only (no SWAP overhead), well within near-term superconducting or trapped-ion coherence budgets – genuinely NISQ-deployable [25, 26], unlike the Givens-butterfly control below. Because the circuit's action is a closed-form composition of real rotations, we implement it as pure PyTorch tensor algebra rather than through a quantum-circuit simulator such as PennyLane [23]; this is numerically identical to what a statevector simulator (or, on the real-amplitude subspace this circuit stays in, real hardware) would produce, and trains roughly 10× faster. We confirmed empirically that gradient norms stay non-vanishing throughout training – no barren-plateau signature – consistent with the circuit's shallow depth and modest qubit count.

2.4 Training Objective and Optimization

QOVT is trained end-to-end with the standard 3-class cross-entropy loss on the [CLS] logits, $\mathscr{L} = -\sum_{c=1}^{3} y_c \log \hat{p}_c$, no auxiliary or quantum-specific loss terms. All circuit angles are ordinary PyTorch leaf parameters and receive gradients through the same backpropagation graph as every other layer – no parameter-shift rule is needed because the RY+CNOT circuit's real-rotation composition is written directly as differentiable tensor operations. Unless noted otherwise (Section 3.5), all experiments use AdamW (lr=3×10^{-4}, weight decay 0.01, β=$(0.9, 0.999)$), a plain cosine learning-rate schedule, 50 epochs, batch size 64, images normalized to $[0,1]$ with no augmentation. We select the checkpoint with the best *validation* AUC (not the last epoch); for the pre-registered round (Section 3.5) the validation split is further divided into a disjoint model-selection half and a held-out test half, with the test half touched exactly once, after checkpoint selection.

2.5 Baselines and Controls

To isolate what the RY+CNOT circuit's quantum, orthogonal, and parameter-count properties each contribute, we compare against three controls that share QOVT's backbone (Section 2.2) but replace the source of U,V:

Givens butterfly (classical, not quantum). U is built as a product of $\log_2 D$ layers of $D/2$ pairwise Givens rotations in a butterfly (FFT-like) pattern – a standard way to parameterize an orthogonal matrix with $O(D\log D)$ angles [5]. This construction does *not* correspond to any quantum gate sequence; we include it purely as a classical ablation on whether the orthogonality constraint alone, independent of any quantum circuit, explains QOVT's behavior. T=192 angles per matrix, 1,536 total.

Sham. U,V are unconstrained $D\times D$ linear maps (`nn.Linear(64,64,bias=False)`) with PyTorch's default Kaiming-uniform initialization – not orthogonal, not structured. T=32,768 total. This is the "no constraint at all" floor.

Matched (rank-1). $U = AB$ with $A \in \mathbb{R}^{D\times r}, B \in \mathbb{R}^{r\times D}$, an unconstrained low-rank factorization with T set to match Givens' parameter count (r chosen so T=1,024), used only in the pre-registered round (Section 3.5) as a parameter-count-matched non-orthogonal control.

Classical MHA. Standard 8-head multi-head attention with independent learned Q,K,V projections, T=$16\times 64^2 =$ 65,536.

Arm	Attention params (T)	Total model params
RY+CNOT (quantum)	384	142,467
Givens butterfly (classical)	1,536	143,619
Matched (rank-1)	1,024	143,107
Sham	32,768	174,851
Classical MHA	65,536	207,619

Table 1: Parameter counts for every arm sharing the QOVT backbone.

3 Experiments

3.1 Setup

We use three DeepLense-style simulated strong-lensing datasets, each a 3-class problem (no substructure / CDM subhalos / axion vortex substructure): Model_I and Model_II (Euclid-like resolution, weaker vs. stronger axion signal respectively) and Model_III (HST-like resolution). Each has $\sim$25,000–29,900 images per class; we use a fixed 9:1 train/validation split (seed 42). All reported numbers are macro-averaged one-vs-rest AUC. Training runs on a single NVIDIA H200 GPU via SLURM.

3.2 Full-Data Results

Arm	Params	Model_I	Model_II	Model_III
RY+CNOT (quantum)	142,467	0.979	0.985	0.995
RY+CNOT sham	174,851	0.980	0.991	0.990
Givens (classical)	143,619	0.981	**0.994**	**0.996**
Givens sham	174,851	0.980	0.991	0.989
Classical MHA	207,619	**0.981**	0.992	0.996
MAE-ViT (reference)	2,722,947	0.978	0.990	0.990

Table 2: Full-data (single seed) macro OVR AUC. All architectures converge within $\sim$1.5 AUC points once given the full dataset.

Given the full dataset, every arm – quantum, classical, and classical baseline – lands within about 1.5 AUC points of each other (Table 2); the interesting regime, and the rest of this paper's focus, is low-data.

3.3 Data-Efficiency Sweep

Arm (T)	N=500	N=3,000	N=5,000	N=8,000	full
RY+CNOT (384)	**0.914**	0.964	0.970	0.966	0.985
RY+CNOT sham	0.881	0.960	0.961	0.965	0.991
Givens (1,536)	0.526	0.939	0.961	0.558	0.994
Classical MHA (65,536)	0.910	**0.964**	**0.968**	**0.980**	0.992

Table 3: Model_II macro OVR AUC vs. training-set size (single seed per cell; the N=500 cell is superseded by the 10-seed Table 5). Givens fails at *both* N=500 and N=8,000 – non-monotonic in N, ruling out simple memorization.

RY+CNOT tracks classical MHA within about 1–2 points at every N we test, despite using 170$\times$ fewer attention parameters. Givens, in contrast, collapses to near-chance (0.526) at N=500, recovers through the middle of the sweep, and then collapses *again* at N=8,000 (0.558) before working at full data. If this were memorization – too few parameters relative to data, or the reverse – we would expect a monotonic trend with N; instead we see the same instability recur at a much larger training-set size, which points to an optimization-landscape pathology specific to the Givens-butterfly parameterization rather than a data-efficiency effect (further discussed in Section 4).

3.4 Circuit Depth Ablation: Testing the Generalization Bound

L_{circ}	T (total, all L=4 layers)	AUC @ N=500	AUC @ full data
4	192	0.896	0.982
8 (default)	384	0.914	**0.985**
12	576	**0.930**	0.981

Table 4: Model_II AUC vs. circuit depth (single seed). Accuracy *increases*, not decreases, with T at low N – the opposite of what the Caro bound (Section 2.1) predicts if it were the operative mechanism.

If the Caro bound were the mechanism behind RY+CNOT's low-data performance, held-out accuracy at N=500 should *fall* as T grows from 192 to 576. It does the opposite (Table 4): more circuit parameters help, not hurt, at low N. We still keep the shallow 8-layer circuit as our default – it is what makes the circuit NISQ-deployable – but we no longer offer the generalization bound as the explanation for why it works.

3.5 Pre-Registered Multi-Seed Confirmation

The single-seed results above are suggestive but not statistically conclusive, and an earlier campaign in this project was retracted after an audit found its "significant" claim rested on a broken sham control and a confirmatory sample that overlapped the discovery sample. To avoid repeating that mistake, we pre-registered this round's hypotheses, primary tests, and Holm-Bonferroni correction procedure in a committed document *before* running any of the training jobs, and disclose every deviation from the main protocol (Section 2.4) here rather than silently: (1) a held-out test split, distinct from the model-selection split, touched once; (2) the addition of the matched rank-1 control; (3) a single uniform learning rate for every arm (no quantum-specific boost); and (4) a different optimization recipe – AdamW, lr=10^{-3}, weight decay 10^{-4}, `CosineAnnealingWarmRestarts`, 30 epochs – used only for this round. We run 5 arms $\times$ 2 datasets (Model_II, Model_III) $\times$ 10 seeds (42–51) at N=500/class, 100 runs total.

Arm (T)	Model_II		Model_III	
	mean AUC	std	mean AUC	std
RY+CNOT (384)	**0.886**	0.070	**0.893**	0.055
Givens (classical, 1,536)	0.508	0.033	0.528	0.047
Matched (rank-1, 1,024)	0.549	0.021	0.570	0.024
Sham (32,768)	0.729	0.153	0.819	0.107
Classical MHA (65,536)	0.749	0.075	0.699	0.119

Table 5: Pre-registered results, N=500/class, n=10 seeds, held-out test AUC. Full statistics in the text.

Primary tests (Holm-corrected, paired one-sided Wilcoxon, 4 tests). RY+CNOT beats matched on both datasets (Model_II Δ=+33.7pp, Model_III Δ=+32.3pp, both 10/10 seeds, both Holm p=0.0039). RY+CNOT beats sham on Model_II (Δ=+15.7pp, 10/10 seeds, Holm p=0.0039, our practical-significance bar) but *not* on Model_III (Δ=+7.4pp, 7/10 seeds, Holm p=0.0137, not significant). We read this as a real but dataset-dependent, unreplicated finding – not a general "quantum beats classical" claim.

Secondary test. Comparing Givens directly against RY+CNOT (two-sided, paired by seed) confirms Givens is worse on both datasets (p=0.0020 each, 0/10 seeds favoring Givens), consistent with the single-seed instability already seen in Table 3. The win is unanimous across all 10 seeds on *both* datasets, unlike the RY+CNOT-vs-sham comparison above, which only reaches unanimity on Model_II – Givens' disadvantage relative to RY+CNOT is the more robust of the two effects in this round.

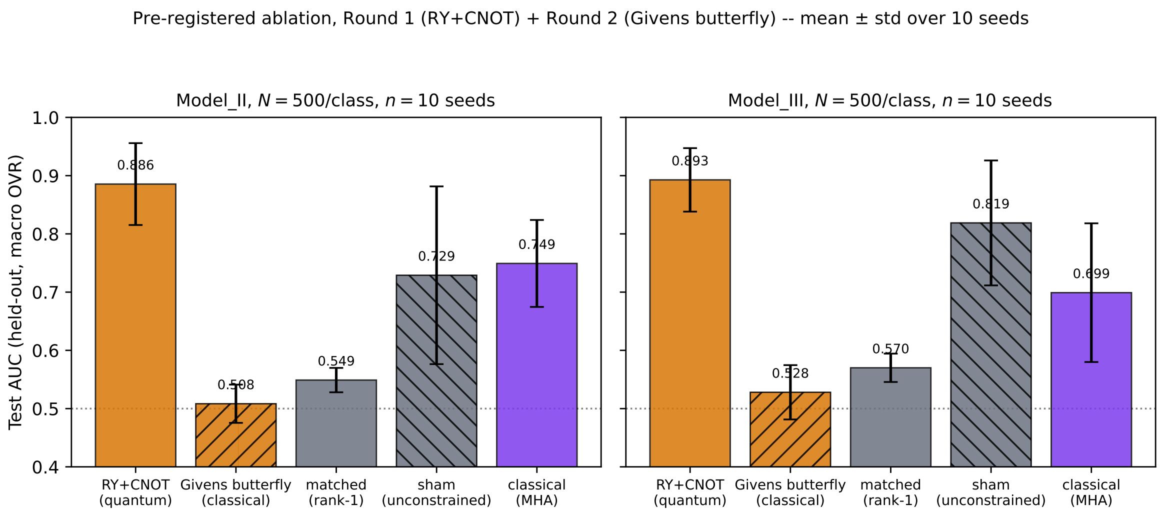

Figure 2: Pre-registered results, mean ± std over 10 seeds, by arm and dataset (Table 5).

4 Analysis

Parameter efficiency comes with spectral stability. RY+CNOT matches classical MHA's accuracy using 170× fewer attention parameters (Table 1) because orthogonality is enforced by construction: $\|U\|_2 = \|V\|_2 = 1$ for every setting of the circuit's angles, so the attention layer's output can never blow up or collapse regardless of T. This is also why RY+CNOT never exhibits Givens' catastrophic failures at any N we test (Table 3) – both circuits are orthogonality-constrained, so orthogonality alone does not explain the difference; the RY+CNOT gate sequence itself is the more stable optimization landscape.

The generalization bound is not the operative mechanism. If the Caro bound explained RY+CNOT's data efficiency, we would expect it to beat classical MHA at low N (where the $O(\sqrt{T/N})$ gap should matter most) and the depth ablation to show *fewer* parameters generalizing better. Neither holds: RY+CNOT ties classical MHA at every N (Table 3), and more circuit parameters improve, not hurt, low-N accuracy (Table 4). We keep the bound as a design heuristic that motivated trying a shallow circuit in the first place, not as an explanation for the results we observe.

Givens' failure is circuit-specific, not generic to orthogonal attention. Givens fails at N=500, recovers by N=3,000–5,000, and fails again at N=8,000 (Table 3) – non-monotonic in N, which rules out a simple memorization or under-parameterization story. RY+CNOT, using less than a quarter of Givens' parameters and the same orthogonality constraint, shows no such instability at any N. The most parsimonious explanation is that the Givens-butterfly parameterization induces a harder, possibly more non-convex optimization landscape than the RY+CNOT gate sequence – an empirical question about the two circuits' loss landscapes that we leave to future work rather than resolve here.

5 Conclusion

RY+CNOT is a genuine, 6-qubit, NISQ-deployable gate-model quantum circuit that produces the orthogonal attention matrices for QOVT and matches a classical multi-head-attention baseline's held-out accuracy at every training-set size we test, using 170× fewer attention parameters and remaining spectrally stable throughout. A pre-registered, held-out-test, 10-seed evaluation shows it significantly beats an unconstrained baseline on one of two datasets – a real, directional, but dataset-dependent and as-yet-unreplicated finding. We find no evidence that the Caro et al. generalization bound explains this behavior, and we show, via a classical (non-quantum) Givens-butterfly control, that constructing "the same" orthogonal matrix without the RY+CNOT gate sequence loses its stability entirely, failing in roughly half of all runs including at 5× more training data.

Limitations. All results are simulated, not run on real quantum hardware. Only the pre-registered round (Section 3.5) uses multiple seeds; the sweep and depth ablation (Sections 3.3, 3.4) are single-seed and should be read as suggestive, not conclusive. The RY+CNOT-vs-sham effect holds on Model_II but not Model_III, and we do not yet know what dataset property drives that difference. Our attention design is asymmetric ($W_Q = I$); we have not tested the symmetric $W_Q = W_K = U$ variant used elsewhere in the literature. We did not log training-accuracy trajectories in enough detail to fully characterize *why* Givens' optimization landscape is harder, only that it is.

Future work. An independent second pre-registered round (a different N, fresh seeds) to test whether the Model_II finding replicates; a symmetric $W_Q = W_K = U$ variant; a loss-landscape or Hessian-conditioning comparison between RY+CNOT and Givens to explain the stability gap directly; a multi-seed rerun of Givens at $N = 8{,}000$ to characterize its failure mode more precisely; and deployment of the RY+CNOT circuit on real NISQ hardware, for which its shallow depth and ring connectivity make it a realistic near-term candidate.

Reproducibility. All training code, the pre-registration document (committed before any of these jobs ran), raw per-seed results, and this manuscript's LaTeX source are public in the project repository.

References

[1] S. Alexander et al. *Deep Learning the Morphology of Dark Matter Substructure*. The Astrophysical Journal, 893(1):15, 2020.

[2] J. Brehmer, S. Mishra-Sharma, J. Hermans, G. Louppe, and K. Cranmer. *Mining for Dark Matter Substructure: Inferring Subhalo Population Properties from Strong Lenses with Machine Learning*. The Astrophysical Journal, 886(1):49, 2019.

[3] M. C. Caro et al. *Generalization in Quantum Machine Learning from Few Training Data*. Nature Communications, 13:4919, 2022.

[4] M. Cerezo, A. Sone, T. Volkoff, L. Cincio, and P. J. Coles. *Cost Function Dependent Barren Plateaus in Shallow Parametrized Quantum Circuits*. Nature Communications, 12:1791, 2021.

[5] E. A. Cherrat, I. Kerenidis, N. Mathur, J. Landman, M. Strahm, and Y. Y. Li. *Quantum Vision Transformers*. Quantum, 8:1265, 2024. arXiv:2209.08167.

[6] M. Comajoan Cara et al. *Quantum Vision Transformers for Quark–Gluon Classification*. MDPI Axioms, 13(5):323, 2024.

[7] N. Dalal and C. S. Kochanek. *Direct Detection of Cold Dark Matter Substructure*. The Astrophysical Journal, 572:25–33, 2002. arXiv:astro-ph/0111456.

[8] A. Dosovitskiy et al. *An Image is Worth 16x16 Words: Transformers for Image Recognition at Scale*. ICLR 2021. arXiv:2010.11929.

[9] Y. Fei, Y. Liu, X. Wei, and M. Chen. *O-ViT: Orthogonal Vision Transformer*. arXiv:2201.12133, 2022.

[10] X. Gao, Z.-Y. Zhang, and L.-M. Duan. *A Quantum Machine Learning Algorithm Based on Generative Models*. Science Advances, 4(12):eaat9004, 2018.

[11] V. Havlíček et al. *Supervised Learning with Quantum-Enhanced Feature Spaces*. Nature, 567(7747):209–212, 2019.

[12] Y. D. Hezaveh, L. P. Levasseur, and P. J. Marshall. *Fast Automated Analysis of Strong Gravitational Lenses with Convolutional Neural Networks*. Nature, 548:555–557, 2017.

[13] L. Hui, J. P. Ostriker, S. Tremaine, and E. Witten. *Ultralight Axion Dark Matter*. Physical Review D, 95(4):043541, 2017. arXiv:1610.08297.

[14] F. Lanusse et al. *CMU DeepLens: Deep Learning for Automatic Image-Based Galaxy–Galaxy Strong Lens Finding*. Monthly Notices of the Royal Astronomical Society, 473(3):3895–3906, 2018.

[15] L. Huang, X. Liu, B. Lang, A. W. Yu, Y. Wang, and B. Li. *Orthogonal Weight Normalization: Solution to Optimization over Multiple Dependent Stiefel Manifolds in Deep Neural Networks*. AAAI, 2018. arXiv:1709.06079.

[16] H.-Y. Huang et al. *Power of Data in Quantum Machine Learning*. Nature Communications, 12:2631, 2021.

[17] I. Kerenidis, J. Landman, and N. Mathur. *Classical and Quantum Algorithms for Orthogonal Neural Networks*. arXiv:2106.07198, 2021.

[18] J. Landman et al. *Quantum Methods for Neural Networks and Application to Medical Image Classification*. Quantum, 6:881, 2022. arXiv:2212.07389.

[19] G. Li, X. Zhao, and X. Wang. *Quantum Self-Attention Neural Networks for Text Classification*. arXiv:2205.05625, 2022.

[20] Y. Liu, S. Arunachalam, and K. Temme. *A Rigorous and Robust Quantum Speed-Up in Supervised Machine Learning*. Nature Physics, 17(9):1013–1017, 2021.

[21] J. R. McClean, S. Boixo, V. N. Smelyanskiy, R. Babbush, and H. Neven. *Barren Plateaus in Quantum Neural Network Training Landscapes*. Nature Communications, 9:4812, 2018.

[22] V. S. Ngairangbam, M. Spannowsky, and M. Takeuchi. *Anomaly Detection in High-Energy Physics Using a Quantum Autoencoder*. Physical Review D, 105:095004, 2022.

[23] V. Bergholm et al. *PennyLane: Automatic Differentiation of Hybrid Quantum-Classical Computations*. arXiv:1811.04968, 2018.

[24] A. A. Prasha et al. *Masked Autoencoder Pretraining on Strong-Lensing Images for Joint Dark-Matter Model Classification and Super-Resolution*. arXiv:2512.06642, 2025.

[25] J. Preskill. *Quantum Computing in the NISQ Era and Beyond*. Quantum, 2:79, 2018.

[26] K. Bharti et al. *Noisy intermediate-scale quantum (NISQ) algorithms*. Reviews of Modern Physics, 94(1):015004, 2022.

[27] S. Sim, P. D. Johnson, and A. Aspuru-Guzik. *Expressibility and Entangling Capability of Parameterized Quantum Circuits for Hybrid Quantum-Classical Algorithms*. Advanced Quantum Technologies, 2(12):1900070, 2019.

[28] K. Terashi et al. *Event Classification with Quantum Machine Learning in High-Energy Physics*. Computing and Software for Big Science, 5:2, 2021.

[29] A. Tesi et al. *Quantum Attention for Vision Transformers in High Energy Physics*. arXiv:2411.13520, 2024.

[30] E. B. Unlu et al. *Hybrid Quantum Vision Transformers for Event Classification in High Energy Physics*. MDPI Axioms, 13(3):187, 2024.

[31] S. Wang, C. Theobalt, and V. Golyanik. *Quantum Visual Fields with Neural Amplitude Encoding*. NeurIPS 2025. arXiv:2508.10900.